School of Mathematical & Physical Science

Research Student Handbook

2015 -2016

http://www.smps.reading.ac.uk/

DISCLAIMER

The Departmental information and the website addresses given in this section of the Handbook are correct at the time of printing. Some of this information may change during the course of your research programme at The University of Reading and you will find *the most up-to-date versions of the documents on the School and Departmental web pages*

Contents

ABSTRACT

This Handbook has three main sections:

General Information: The nature of the School, its constituent components, and its relation to the rest of the University. This section gives details on the layout of the Departments, their facilities, and the responsibilities of the individual staff members. It also gives details of student representation within the School and University, travel funding, careers and professional societies.

Training: options available in the different departments and how training programmes are tailored to individual students.

Monitoring and Progression: The School procedure for monitoring research students to ensure that they are making satisfactory progress. The process is two-way and enables students to ensure that they are receiving appropriate levels of supervision and support. Information is given on the stages in formal progression of your PhD and on extension and suspension in PhD funding and registration.

In addition to this handbook you will find a wealth of information available on the School PhD webpage (http://www.smps.reading.ac.uk/intranet/home/phd_information/), the University Graduate School webpages (http://www.reading.ac.uk/graduateschool/gs-home.aspx) and the Examinations Office webpages (http://www.reading.ac.uk/exams/).

The Graduate School webpage includes a link to the University's Policies and Procedures for Research Students which outlines a minimum set of requirements for all parties involved in the awarding of postgraduate research degrees at the University of Reading and includes lists of the responsibilities of the student and supervisors in the PhD process: http://www.reading.ac.uk/graduateschool/currentstudents/gs-policies-and-procedures.aspx

For International students, there is a useful section on the student pages at http://www.reading.ac.uk/internal/student/international-students/ .

At the end of this handbook an annex describes how UK research councils use the data they hold on students funded by them(this comes from a briefing note provided by the research councils).

1. GENERAL INFORMATION

1.1 Overview of the School of Mathematical and Physical Sciences

The interdisciplinary School of Mathematics, Meteorology and Physics was formed on the 1 August 2004. The School consists of the Department of Mathematics and Statistics, the Department of Meteorology and the Statistical Services Centre (SSC). The Computer Science Department is expected to join the school in 2016. The SSC is a self-financing unit which provides statistical training as well as statistical consultancy to clients from inside and outside the University.

Natural Environment Research Council (NERC) distributed centres include the *National Centre for Atmospheric Science* (NCAS) and *National Centre for Earth Observation* (NCEO). NCAS has three divisions: climate, weather and composition. The Department of Meteorology hosts the NCAS Climate Directorate. Some NCAS-weather and NCEO staff are employed in the Department of Meteorology and in the School.

Meteorology hosts several research groups from the Met Office, as part of the Met Office Academic Partnership (MOAP). The other MOAP universities are Leeds, Exeter and Oxford.

All of these Departments and Research Centres have affiliated research students. Hence students may be members of the departments of Mathematics and Statistics or Meteorology, and additionally be attached to NCAS or NCEO. Although in general the guidance given to students is similar in the different departments, some differences in procedure exist.

The *Department of Mathematics and Statistics* (http://www.reading.ac.uk/maths-and-stats/) offers undergraduate BSc and MMath programmes in both pure and applied mathematics, statistics and applied statistics as well as joint degrees with economics, meteorology, computer science and psychology and the Mathematics of Planet Earth Centre for Doctoral Training (MPE CDT) in conjunction with Imperial College London. The department has an active postgraduate research programme including a PhD in Mathematics and Statistics. The department has a strong research record in Applied Analysis, Numerical Mathematics (including Conservation Laws and Data Assimilation), and Pure Mathematics. The research in Statistics focuses on biostatistics including clinical trials, meta-analysis, capture-recapture modelling, statistical genetics and Bayesian statistics. The Department is located on the Pepper Lane (west) side of the lake bisecting the main Whiteknights campus (building 4 on the University campus map), whereas the Mathematics of Planet Earth Centre for Doctoral Training resides in the nearby Lyle building (building 32). The SSC is located in the Harry Pitt building at Earley Gate (building 56).

The Meteorology Department (http://www.met.reading.ac.uk) is world-leading centre of excellence in the atmospheric sciences. In 2006 Meteorology at Reading was awarded the Queen's Anniversary Prize for Higher and Further Education. It is the only UK Department to offer a full range of undergraduate and postgraduate courses in the atmospheric sciences. Meteorology runs MSc courses in *Applied Meteorology*, *Applied Meteorology and Climate with Management*, *Atmosphere, Oceans & Climate* and *Data Assimilation & Inverse Problems in Geophysics*. The Meteorology Department is located within the Meteorology (building 58 on the campus map), Harry Pitt (building 56), Agriculture (building 59) and Lyle Building (building 32). The SCENARIO (Postgraduate centre in the SCience of the Environment: Natural and Anthropogenic pRocesses, Impacts and Opportunities) Doctoral Training Partnership occupies the 2nd floor of the Lyle building together with the Mathematics of Planet Earth Centre for Doctoral Training.

1.2 Key people in the School and Departments

The following are key people with whom you're likely to interact initially (see local www pages for more information).

School

Professor Simon Chandler-Wilde:	*Head of School (based in Meteorology) (Until December 2015)*
Mrs Marguerite Gascoine	*School Manager*
Mrs Louise Barrow	*Executive assistant to Head of School*
Mrs Chantal Murfitt	*Executive assistant to Head of School*
Mrs Jean Teall	*PA to Head of School*
Dr Karen Ayres (Statistics):	*School Director of Teaching and Learning*
Sophie King-Warning	*Special Needs Co-ordinator*
Professor Suzanne Gray (Meteorology)	*School Director of Postgraduate Research*
Mr Ross Reynolds (Meteorology)	*School Director of Internationalisation*
Professor Beatrice Pelloni	*Deputy Director of the Mathematics of Planet Earth Centre for Doctoral Training*
Professor John Methven	*Director the SCENARIO Doctoral Training Partnership*
Ms Jill Hazleton	***Doctoral Training Centre Administrator***
Ms Linda Watson	*School MSc Administrator*

Mathematics and Statistics

Dr Stephen Langdon	*Head of Department (Research, Teaching and Learning)*
Professor Alexei Likhtman	*Departmental Director of Postgraduate Research Studies: Mathematics*
Dr Fazil Baksh	*PhD in Statistics Coordinator*
Mrs Peta King	***Postgraduate by Research Administrator:*** *Mathematics and Statistics*
Dr Amos Lawless	*Library Representative*

Meteorology

Vacant	*Head of Department (Academic/Res. staff)*
Professor R Giles Harrison	*Head of Department (External Affairs)*
Professor Keith Haines	*Head of Department (Research/Building)*
Dr Maarten Ambaum	*Head of Department (Finance/Support staff)*
Mrs Dawn Turner	*Executive Assistant to the Heads of Department and External Affairs Co-ordinator*
Mrs Debbie Turner	*PA to Heads of Department*
Professor Sue Grimmond	*Departmental Director of Postgraduate Research Studies: Meteorology*
Mrs Christine Macfarlane	***Postgraduate by Research Administrator:*** *Meteorology*
Dr Nicolas Bellouin	*PhD Admissions Tutor*
Prof Andrew Charlton-Perez	*Meteorology MSc Director*
Dr Tom Frame	*Meteorology MSc Admissions Director*
Miss Kim Oakley	*Harry Pitt Administration, Safety Officer Harry Pitt*
Mr. Dan Bretherton	*Research HPC manager*
Mr Neil Blanchonnet	*Senior IT Business Partner*
Mr Stephen Gill	*Health and Safety Coordinator, Building and Facilities Manager*
Mrs Catherine Turner	*Librarian*
Ms Carly Wright	*Chair of Meteorology Postgraduate Research Forum*

1.3 Facilities

The various school buildings are open between the normal working hours of 8 am and 6 pm each working day. All research students are issued with a swipe card that will enable them to gain access outside these hours and at weekends/holidays. If you are in a Department outside working hours then you must sign in (and sign out) in the *Building Occupants Register Book in the entrance lobby*. Please ensure that you also sign this book if you are in a Department but continue working after 6 pm. The relevant Departmental safety code can be found in each office. You should read this as all staff and students are required to follow this code.

Mathematics and Statistics

General facilities

PhD students have their own allocated **desk space with their own computer facilities**. The Mathematics **Common Room**, Room M112, is available for use by all staff and students and visitors to the department. It has a selection of easy chairs and whiteboards, and is the place for much informal contact and personal discussions that may disturb the people working in your shared office. Students may make their own tea/coffee in the kitchen on payment of the correct fee into the jar in the kitchen.

In general the availability of office equipment and similar services should be checked with the Postgraduate Administrator, Mrs. Peta King, in Room M209. Reproduction of printed matter may be carried out on the **photocopier** in the department (subject to the usual copyright restrictions). Each student will be given a code number for use of this facility. Although their use of the photocopier is monitored, PhD students are not charged for reasonable use of the facility.

Mail for students is placed in a named post-tray in the Post Room (room M214). **It is important that you regularly check for mail since urgent and often important information on course arrangements is to be found here**.
There is a large **departmental postgraduate library**, which contains many books of interest to postgraduate students. Books may be borrowed for unlimited periods but **must be signed for** and, for the convenience of others who may wish to consult them, they should normally be kept in the department.

Computing

The Department of Mathematics & Statistics owns restricted-access PCs, mainly for postgraduate use. Unlike the ITS facilities, these workstations have restricted login access. They may be used by all staff, research and postgraduate users in the department but not by undergraduates or other members of the University. The printing facilities connected to them are likewise restricted. In most other respects the departmental machines are identical to the ITS lab machines. The rooms containing these machines have combination locks (the combination may be obtained from Mrs Ros Young in M203) and for security reasons should be locked when unoccupied, even during normal working hours.

The department is getting new printers installed and these will be available in rooms JJT 222, JJT 206, Maths 114, Maths photocopy room, Maths 316 & Maths 313 and will all be accessible via campus card. At the moment there is no charge for normal use of these printers; however their use is recorded and monitored and quotas may be implemented if unreasonable use occurs. (For example they should not be used in place of the photocopier.)

The PCs use shared common file space for users-files. This has the advantage that user's files, emails etc. are available at any PC. However, file space is limited and so should not be used to excess – in particular users should not store large amounts of email messages or nonacademic material such as pictures, games or audio files.
Consideration for other users should always be made. For example, users should not deny others access to a PC by leaving a console logged in but unattended for periods over 5 minutes or by using lockscreen programs. File space is short so users should compress files not in frequent use, remove executables which can easily be recompiled and not store nonacademic material such as pictures, games or audio files.

Most offices have a **phone** which can be used for local calls and calls to specified non-local numbers. Please note that the use of all phones in the Department is monitored and questions will be asked if use seems excessive.

Meteorology, Lyle, Harry Pitt and Agriculture Buildings

General facilities

The Meteorology Building has a large **common room** used by all students and staff. Coffee and tea can be obtained from the kitchen area on payment of a small fee. The money raised by the coffee fund in the Meteorology and Lyle buildings is used to support Departmental activities such as the annual Christmas party and summer barn dance and BBQ. Microwaves and an expresso coffee machine are also available for use. The kitchen area is maintained by research students. The Lyle Building also has kitchen areas to make coffee/tea etc. The Agriculture Building has its own café where lunch and snacks can be purchased. Harry Pitt first floor has three communal rooms: the conference room (176), the seminar room (175) and the Tap Room or kitchen. On the first floor there are toilets just outside the south entrance and along the Applied Statistics north corridors on both first and second floors.

The Meteorology Building has a well-stocked **library** from which books can be borrowed. Most of the journals commonly required are available online (other Meteorological journals are available at the *University library*). Journals *may not be removed* from the library and so you should photocopy any articles you wish to read away from the library area. Research students may use the photocopiers in 1L37 and 1U24.

All students are allocated a desk and a computer (usually a thin client terminal or similar which can be used to access multiple-use servers) in one of 4 PhD offices in the Meteorology Building (each suitable for about ten students), in the Harry Pitt Building (179 & 288), or on the 2nd, 3rd or 5th floors of the Lyle Building. Access to the Harry Pitt Building is normally through the north entrance. Alarm codes are required to access Harry Pitt building in the morning (the first person in on each floor disables the alarm for that floor) and also out of hours.

Each office has a ***phone*** which can be used for local calls and calls to specified non-local numbers (e.g. the Met Office in Exeter). Please note that the use of all phones in the Department is monitored and questions will be asked if use seems excessive.

Meeting and seminar rooms can be booked by students for meetings. Please check room availability on the internal department webpage (http://www.met.reading.ac.uk/intranet/home/services/room-booking/ and then email room-booking@met to reserve the room. In Meteorology the porter will deliver mail to your desk. In the Lyle building the administrator will distribute to your desk.

Personal copies of periodicals are usually left in the coffee room for collection. It is important to tell the porter, the secretaries in the Postgraduate office, and the computing staff if you change desks as records are kept of who sits where. The School Postgraduate Office is located in room 1L42 of the Meteorology Building, near the coffee room.

Computing

Most research students use the UNIX/LINUX network facilities available in the Department (which includes your "home" disk space with back-up, laser printing and internet connection to the outside world). Windows-based PCs are also available in the computer rooms (although a Windows emulator is available via the network). For intensive processes students use a compute cluster called the "Met-Cluster", or remote access to supercomputers outside the University. Students receive an introduction to Departmental computing as part of their induction. If you have a computing problem then it is worth asking other students if they know how to fix it and looking at the itmet web pages at http://www.met.reading.ac.uk/it/home. If this doesn't help then the problem should be reported using the online Self Service Desk (https://uor.topdesk.net/tas/public/) or by email to its-help@reading.ac.uk.

The Department has a met-social e-mail alias to which all new students are subscribed. This is a good source of information about the various social and sporting activities that occur in the Department. There are also other e-mail lists to which you can subscribe, mainly specific to particular computer languages or software (e.g. python, matlab).

Much of the information you will require about the Departmental computing and library facilities (including a search facility for library books and journals) is available online: see http://www.met.reading.ac.uk/it/home

1.4 Student Representation

Mathematics and Statistics
The Department has a Postgraduate Forum consisting of one representative from each year of Research students, together with the Head of Department and the Director of Postgraduate students.

Meteorology
The Postgraduate Research Forum consists of elected representatives from each current year of the Postgraduate Research student body with a chairperson and international student representative, together with the Department Head of Research and the Departmental Director of Postgraduate Research studies. Its meetings are convened by a Student Chairperson and the minutes of its meetings are available for all Postgraduate Research students to read. The Forum considers all matters which affect the academic life of our Postgraduate Research by Research students, including administrative arrangements, training programmes, provision of Departmental facilities etc., and it is an important channel of communication between the students and academic staff. The Forum meets once each term. If you have any concerns that you think the Forum should consider, please discuss them with one of the Forum members.

University
A PhD student from each Department is nominated to represent postgraduate research students at a University level. Meetings of the representatives are coordinated by the Graduate School.

1.5 Travel and Conferences

You should discuss business travel to meetings and conferences with your supervisor(s) and together decide what meetings would be useful to participate in. Note that students are covered by the University's **business travel insurance policy** for overnight stays (http://www.reading.ac.uk/internal/finance/TravelandInsurance/fcs-ins-business.aspx). Before travel you are required to fill out a short **online risk management form** for students that is available on the web link above.
You are also required to book most travel through the university's nominated travel agent (currently CTM) which can be done directly and paid for by any of the Administrative team.
If your PhD involves **field or laboratory work** you will need to complete **a risk assessment form** for the activities in with your supervisor and submit it to the appropriate Department Health and Safety Coordinator. For Meteorology, health and safety information and forms can be found at http://www.met.reading.ac.uk/intranet/home/health/travel.html .

Important note: Non-EU students are **required by the UK Border Agency (UKBA) to notify the University when they plan to spend time away from Reading**. Please tell the Postgraduate Administrator (for Met or Mathematics & Statistics as appropriate) the dates when you will be absent from the UK before you leave on a trip for any reason, not just work. Also, please tell us the dates of trips elsewhere in the UK for prolonged periods (more than a week). The University is responsible for recording this information for the UK Government during your stay in Britain and it is a condition of your visa for entry into the UK.

All students should notify their supervisor when they plan to spend prolonged periods (*more than a week*) away from Reading. Please note that if you are enrolled as a full-time student you are expected to be working all year (not just term time) as you would in a full-time job and you are entitled to the same duration of holiday as University employees. You are strongly advised to arrange your holidays to be out of term time since otherwise you are likely to miss useful seminars or other events and supervisors are likely to be more available during term time (as they often travel to conferences or take their own holidays out of term time). ***Failure to maintain contact with your supervisor***, and inform them of your absences, could lead the School instigating procedures for terminating your registration on the grounds of lack of academic engagement.

Mathematics and Statistics

It is expected that research students (PhD) attend one national and one international conference during their period of study. Students supported by EPSRC and industry have an allowance for this built into their funding (in addition to the stipend). The Department will usually contribute funding for limited conference attendance for other students. For statistics, a natural choice of the national conference would be the annual meeting of the Royal Statistical Society where there are several events available specifically tailored for early career statisticians.

Flights and hotel bookings need to be booked through the university's approved travel company. Conference registration fees that you need to pay before your trip may be paid for using the Departmental University credit card. This should be arranged via the Administration team. You should ensure that all receipts from your trip are kept as you may need these to claim back money spent on food/travel etc. on your return. Claims forms are available online and should, when completed by you and signed by your supervisor, be returned to Ros Young in M203 for signing by the department representative.

Meteorology

A few students (such as NERC-project students) have their own travel grant held by their supervisors. For these students, all travel and subsistence costs are obtained directly from their grant. Students funded by the SCENARIO NERC Doctoral Training Partnership should speak with the DTP Administrator (*Jill Hazleton* **-** j.hazleton@reading.ac.uk) regarding funding and expenses. All other students who do not have their own separate source of travel funding (including NERC funded Doctoral Training Grant, NERC Industrial CASE students and some overseas students) obtain funding for travel to meetings via the Department. However, the funding limits and procedures are broadly similar independent of funding source.

Students funded by the **SCENARIO NERC Doctoral Training Partnership** should speak with the DTP Administrator (*Jill Hazleton* **-** j.hazleton@reading.ac.uk) if they have any queries regarding funding and expenses. Each student is allocated £2000 for travel and subsistence during the period of their research. This money is only available for travel to conferences and meetings and unused funds cannot be accessed for other purposes. All your expense claims should go through the DTP Administrator. Students may exceptionally exceed their £2000 cap if a case is made to the SCENARIO DTP Director.

Students who **do not have their own separate source of travel funding** (including NERC funded Doctoral Training Grant and NERC Industrial CASE students and some overseas students) obtain funding for travel to meetings involving an overnight stay in the UK or abroad via the Department. Each student is allocated £2000 for travel and subsistence during the period of their research. This money is only available for travel to conferences and meetings and unused funds cannot be accessed for other purposes. Expenses associated with fieldwork and travel need to be budgeted and agreed in advance (please see the travel approval form on web page http://www.met.reading.ac.uk/intranet/student/phd/). Students, in consultation with their supervisors, should draw from their allocation to pay for all meetings which involve an overnight stay. Students can obtain the balance remaining of their allocation from the Department finance administrator, Zala Hussain (a.hussain@reading.ac.uk). Students funded by the NERC Doctoral Training Grant may exceptionally exceed their £2000 cap if a case is made to the Departmental Director of Postgraduate Research Studies and may exceptionally use the money to pay for publication charges (for publishing academic papers) if agreement is sought **prior** to submission of the manuscript.

The Department will additionally fund attendance at one-day meetings (e.g., RMetSoc Wednesday meetings). Students are encouraged to attend one UK meeting of more than one day in each year and one major international conference during their PhD (usually during the 3rd year of research). Students should also seek support for conference attendance from other sources where possible (this may allow you to attend more meetings than your Departmental allocation of funding would permit). For example, many major international conferences have funds available for students to attend and the Royal Meteorological Society also provides some funding for members.

Students are also encouraged to attend one summer school (usually at the end of the 1st year). Supervisors are encouraged to request funding for students to attend summer schools from the funding body prior to the arrival of the student. The Department recognises that this is not always possible therefore, the Department has some money reserved for research student attendance at summer schools (up to £750 per student). Requests for this funding should also be made to the Departmental Director of Postgraduate Research Studies or DTP Administrator as appropriate.

Flights, hotel deposits and registration fees that you need to pay before your trip may be paid for using the Departmental University credit card. This should be arranged via *Christine Macfarlane* (c.m.macfarlane@reading.ac.uk) in the Postgraduate Office in Meteorology, your supervisor's PA or the SCENARIO administrator (*Jill Hazleton* **-** j.hazleton@reading.ac.uk) as applicable. You should ensure that **all receipts from your trip are kept** as you will need these to claim back money spent on food/travel etc. on your return. Claims forms are available from an administrator and should be completed electronically, printed off and signed by your supervisor. They then need to be handed in to an administrator for signing by the Department representative. If necessary you can apply for a cash advance to cover subsistence and accommodation costs before you leave.

1.6 Getting a Job/Networking

The typical destinations of postgraduates leaving the School of Mathematical & Physical Sciences are postdoctoral/teaching positions in universities, research positions in government institutes (e.g. Met Office), and science positions in industry (e.g. environmental science, remote sensing, computing, and engineering). Relevant job opportunities get advertised on Departmental www pages occasionally (see also the met-jobs mailing list at http://www.lists.reading.ac.uk/mailman/listinfo/met-jobs for meteorology-related positions). Your supervisor will be very willing to give advice and to provide references for jobs. It is a good idea to ask/forewarn beforehand. Part of the postgraduate experience involves getting to know who is working in your field and communicating with them. If your topic forms part of a wider project then you have an obvious means of developing such a network. Seminars and conferences are also good places to initiate professional relationships and get yourself known. We encourage this and provided the result is positive it can also help in getting jobs!

1.7 Professional Societies

<u>Meteorology</u>
We encourage you to become a student member of the Royal Meteorological Society and to attend the regular monthly meetings of the Society. Aside from other benefits, the Society has funds available for its members to support scientific visits, conferences etc.

<u>Mathematics and Statistics</u>
There are societies available to join such as the London Mathematical Society or Royal Statistical Society. Aside from other benefits like access to journals, societies often have funds available for its members to support scientific visits, conferences etc.

1.8 Welfare

We do our best to ensure that you will have a productive and enjoyable experience as a postgraduate student. If you need support, please don't just 'grin and bear it' as there is usually something we can do to help. You can talk to someone that you feel comfortable talking to. This might include fellow students, your supervisors, the postgraduate administrator, or your Departmental (or School) Director of Postgraduate Research Studies. There is a lot of support available through Student Services Centre and the Counselling and Wellbeing Service in the Carrington Building and the Student Union (RUSU). There is information on welfare available on the University web pages – see http://www.reading.ac.uk/internal/student/student-advice/stdserv-student-advice.aspx. The University also has Medical Practice and Peer Support Service. Some issues will be appropriate to raise at your Departmental Postgraduate research forum (mainly if they affect several students); if so, please contact the Chair of this forum for your Department.

1.9 Diversity and Inclusion

By joining the School of Mathematical and Physical Sciences you are becoming a member of a community which believes in the principles of equality and inclusion. We actively strive to become an ever more inclusive workplace and a learner community in which diversity is rightly recognised as an asset. As a key School in a research intensive University we value diversity because it enriches the pool of talent we can bring to our research and teaching endeavours as well as our day-to-day operational business. The School aims to ensure that all its working practices enable individuals to succeed independently of their gender, ethnicity, sexual orientation, disability or any other protected characteristic as defined in the Equality Act (2010).

The School is proud to hold an Athena SWAN Silver award for our work on promoting gender equality and the University itself holds an Athena SWAN bronze award and is a Stonewall Diversity Champion of LGBT+ rights. In addition the University also participated in the pilot for the Equality Challenge Unit's Race Equality Charter. If you are interested in participating in our School's work on promoting equality then please contact either of the Chairs of our Equality and Diversity Committee (details below).

Equality and Diversity Committee (chairs): Dr Joy Singarayer (j.s.singarayer@reading.ac.uk); Dr Calvin James Smith (calvin.smith@reading.ac.uk)

1.10 Special Needs

Sophie King-Waring (ext 7612) is the School Special Needs co-ordinator. It is her role to ensure that information on student disabilities (including disabilities such as dyslexia) is communicated to the appropriate staff within the School. She also communicates with the examinations office if necessary.

2. TRAINING

2.1 Graduate School (RRDP) and Generic Training Courses

The University Graduate School coordinates postgraduate researcher training across the University. It is also host to the Doctoral Research Office (DRO) who deal with all matters regarding registration, funding and progression of PhD studentships (www.reading.ac.uk/graduateschool/gs-home.aspx).

The Reading Researcher Development Programme (RRDP) has courses on, for example, how to write a thesis, publish papers, and interact successfully with your supervisor. It is *your responsibility to enrol on these courses*. **You are expected to participate in five of these courses (or similar generic training courses offered by the School/University) in your first year, three in the second year and three in your third year (adjusted as required for part time students).** Attendance at the courses is registered and treated as part of your progression requirement by the University and some funding bodies (e.g., all Research Councils UK).

2.2 Useful References

Some books and www sites on doing a PhD that students have found useful are given below:

Evans D (2011) *How to write a better thesis*, Melbourne University Press (378.242-EVA in University Library)
Greenfield T (2002) *Research Methods for Postgraduates* (2nd Edition)
Phillips EM and DS Pugh (2012) *How to get a PhD: a handbook for students and their supervisors* (6th Edition), Open University Press, Buckingham.
Wolfe J *How to write a PhD thesis*, www.phys.unsw.edu.au/~jw/thesis.html (last accessed 17/8/15)

2.3 Induction

During your first week at Reading (usually the week before the Autumn term starts) you will be involved in registering for your course and supporting university services. The University Graduate School also provides an induction meeting for new postgraduate research students several times per year at which students will receive **a welcome pack** and details of the Reading Researcher Development Programme (RRDP).

Departmental induction programmes:

Mathematics and Statistics
The Departmental Induction usually takes the form a gathering in the Common Room followed by a talk from the Director of Postgraduate Research Studies and a tour around the University and the Department. This gives you some guidance on your new role as a research student and addresses some of the issues which may arise. The Department introduces you to Staff and other postgraduate students. We familiarize you with the facilities and opportunities available.

Meteorology
The Department arranges an Induction Programme during the week preceding the start of the Autumn Term for new research students. The programme complements the University's welcome programmes. It is intended to introduce you to the Department and to each other. An important component is to help you to make the transition to being a research student, and to raise some of the issues faced by those making this transition. You will also go on a tour of the Department and its facilities, including a Health and Safety briefing. Students funded by the NERC SCENARIO Doctoral Training Partnership will have additional induction sessions attended by all students funded through this route (not just those registered in Meteorology).

We also operate a *buddy system*, whereby each new research student is allocated a buddy from the 2nd or 3rd (or occasionally 4th) year research students. The student chair of the Postgraduate Research Forum allocates buddies for the new students. Your buddy provides a contact to answer questions about the Department, to help you to integrate with the rest of the student body and to help you find any advice you may need about any aspect of your Postgraduate Research studies. If you start in the autumn term then you will meet your buddy at lunch on the first day of 'Welcome week'. A *3rd year poster conference* is also held during Welcome week and **all new PhD students are required to attend.** This is an opportunity for our current 3rd year PhD students to present a picture of the science that is being investigated within the department.

2.4 Overview of Taught Training Courses

Different students have different needs for taught courses during their postgraduate career, depending largely on their competency to undertake their research topic and their intellectual motivation. You are encouraged to complete a Learning Needs Analysis with your supervisors to assess your current skillset and determine where there are gaps – a link to this can be found here: http://www.reading.ac.uk/graduateschool/supervisorsandresearchstaff/gs-staff-policies-and-procedures.aspx However, the School does not require you to send the completed form to a Departmental of School Director or Postgraduate Research Studies.

There are two main types of such courses:

Taught Courses at the University
In addition to generic training provided by the Reading Researcher Development Programme (RRDP), PhD students take subject-specific training in the form of MSc modules, mainly given within the School and other Schools. There are also courses given at different institutions with video link access, which can be followed from Reading and will count as internal courses. In particular, Reading is an associate member SEPnet (the South East Physics Network http://www.sepnet.ac.uk/index.html) which provides some of these courses.

Taught Courses outside the University
If the University of Reading does not provide the appropriate course then students may go to other universities, institutions, research councils, or commercial companies to be taught. Examples of this may be a short course on the use of a computer package, or a European or international scheme to provide mixed nation training. In particular there are short courses sponsored by NERC, every one or two years, that are often useful. The most relevant are
Vitae - UK Grad Schools Programme
NCAS Climate Modelling summer school
NCAS Summer School in Atmospheric Measurement
NERC Earth Systems Science summer school

Details of these can be found at:
http://www.nerc.ac.uk/funding/available/postgrad/advanced/

Important note: It is in your best interests not to overburden yourself with too much coursework at the expense of getting your teeth into your own topic.

2.5 Subject-specific Training (MSc Modules)

The School runs several Masters Courses. PhD students are required to attend some modules depending upon their prior experience. *You need to discuss this with your supervisor* before or upon arrival.

Meteorology Department

PhD students registered in the Meteorology Department typically take modules from the MSc Atmosphere, Oceans and Climate (www.met.reading.ac.uk/intranet/student/msc) programme. They can also access appropriate modules from other degree programmes run by the Department or, more rarely, by other Departments either within the University or, dependent on studentship funding, from partner Universities (e.g. students funded by the SCENARIO Doctoral Training Partnership can access modules from the University of Surrey).

The key modules from the autumn term (see http://www.reading.ac.uk/module/) that PhD students in Meteorology often attend are

- *Introduction to weather systems (MTMG01)*
- *Atmospheric Physics (MTMG02)*
- *Fluid Dynamics of the Atmosphere and Oceans (MTMW98)*
- *Introduction to Numerical Modelling (MTMW12)*

There is a wide selection of modules for the spring term (see http://www.reading.ac.uk/module/). The core module, also attended jointly by students from Meteorology and Maths, is

- *Numerical Modelling of the Atmosphere and Oceans (MTMW14)*

The MSc Data Assimilation and Inverse Modelling in Geosciences is jointly run by Mathematics & Statistics and Meteorology Department and it also has a wide selection of modules available.

Often students starting postgraduate research in Meteorology have little or no previous knowledge of fluid mechanics or meteorology. It is the expectation most students will take three modules in one of the Meteorology MSc courses in the autumn term and the three most appropriate advanced modules in the spring term (see http://www.reading.ac.uk/module/). Students who have a meteorological background probably only need to familiarize themselves with the course syllabus. Occasionally it may be more appropriate for students to attend courses from other departments or the MAGIC modules (see below).

Module assessments (including examinations) should be completed in order to gauge your progress. Exemptions from module attendance and/or assessment can be obtained if you have previously completed a similar BSc or MSc module. A *written case for exemption from courses should be made by the supervisor to the Director of Postgraduate Research Studies before you first register*. *Students should speak to their supervisor in the first instance if they believe they should be exempted.* Students who are not exempt are asked to submit a list of module choices and the basis for their choice.

It is expected that a good pass be achieved in the MSc exams (in particular the advanced modules) in order to permit you to proceed to the second year. A good pass is frequently taken to be a mark of **around 60%** on written papers. However, the Postgraduate Research Board of Studies reserves the right to exercise discretion in the interpretation of "good pass". This is in order to make due allowance for the very varied background of students who undertake research in our Department.

Although some students start their PhD after the beginning of the Academic Year, they are still required to attend the full programme of taught courses. This may mean sitting some examinations in the second year. If possible, it is better to gain a basic grounding in the subject early in your time at Reading, leaving you free to concentrate on your research later in your time. The Department reserves the right to postpone your registration until the following October, rather than a mid-year start, if it is appropriate to do so.

Mathematics and Statistics

Students starting a Postgraduate Research in Mathematics or Statistics are usually required to take **5 modules in their first two years of studies**. These modules need to be agreed with the supervisor, and approved by the Director of Postgraduate Research Studies. The result from the first module that you take will be discussed at the first monitoring session.

The modules can be selected among the taught MSc courses, or the few specialized term- or week-courses offered every year by the department. In addition, the department of Mathematics is part of MAGIC, a UK-wide consortium of 18 universities, offering taught courses for student on a Postgraduate Research by Research programme. These course are available to Reading PhD students in videolink. For the list of courses available in this academic year, please see http://maths-magic.ac.uk/courses.php

Please note: EPSRC funded PhD students are required to complete 100 hrs of assessed work in the first two years.

Possible other courses also include Communicating Mathematics, instruction in LaTeX, Fortran, C++, Matlab and Maple.

Language skills

Students whose first language is not English may find themselves at a disadvantage. We may ask such students to attend additional English language classes if we conclude that their performance is being adversely affected by their language standard. More details on the *University's English Language support programme* can be found here: http://www.reading.ac.uk/ISLI/in-sessional/islc-in-sessional-introduction.aspx

2.6 Transferable skills

Oral Presentations

The ability to communicate the results of your research, both verbally and in written form, is an important skill that we aim to develop during your time in Reading. It is a University requirement **that students must give at least one oral presentation in each year.** The ways in which this is achieved vary between the different departments.

Mathematics and Statistics: In May of their first year students present a 40 minute **literature seminar** to staff and other first year postgraduates in the department. This will be in an area related to their research and may include some of their own work. If the student started at a date different from October, this seminar takes place at some appropriately shifted date.

During your second year you will also present a 50 minute seminar on their work, in the regular seminar series. During your 3rd third year you will give another presentation of your work. For students whose research is in the area of numerical analysis, this will usually take the form of a talk to an inter-University group at the Universities Joint Research Symposium. The Department will encourage you to present your work at appropriate scientific conferences and meetings. We also encourage students to write up their results for publication in international journals.

Meteorology: We expect all *second year* students to participate in an annual mini-conference *Quo Vadis* in which they give a short presentation to members of the Department on their research. The students also run an informal weekly meeting, called *Research Student Club*, where you have the chance to practice presenting your research to an audience. Students entering their *third year* are expected to present in a *conference-style poster* session attended by staff and arriving first year students during Welcome week. Please note the dates of Quo Vadis and the poster session in your diaries as soon as they are announced as it is important that you attend these events. We also expect all students to give a *50 minute lunchtime seminar* later in their research, and the Department will encourage you to present your work at appropriate scientific conferences and meetings. We also encourage students to write up their results for *publication in international journals.*

Publication of Research Results

We encourage students to publish their research results in quality journals. Sometimes this only happens after the thesis is complete. However, getting a paper published before the examination gives you the confidence that one of the criteria for awarding the degree, that your work is publishable, has been proven. It also improves the C.V. you submit for your next job. The process of organising, writing, submitting and revising a manuscript is an education in itself. Your supervisor should be able to give substantial help here. Many journals require you to pay publication charges (or 'page charges') to publish your work. Please ensure that you discuss the source of funding for these with your supervisor prior to submitting your paper (preferably before deciding on the journal for which your manuscript is targeted). The Departments do not have a nominated pot of money to fund student page charges although there may exceptionally be funding available for page charges for students funded by the NERC Doctoral Training Grant or Doctoral Training Partnership if agreement is sought **prior** to submission of the manuscript.

Seminars and meetings

Students are required and expected to attend the *regular seminar series* in their Departments, together with any specially organised research events as advised by their supervisor, throughout their time at Reading. Although initially it is likely that students will not fully comprehend the whole of these seminars, which may not be in the specific area of their research, it is still *essential training for the communication of scientific ideas* and provides a broadening of experience. ***It is far too easy to become embroiled in the narrow area of one's own research and to be unaware of other areas.*** Students are expected to have a *general knowledge of their field* and this *can be assessed during the oral examination of the thesis.* Knowledge of other areas of research, outside your specialised area, will also help you to determine possible areas for your future career.

Mathematics and Statistics

There are several seminar series in Mathematics during term time. Some of these are given by staff, research assistants and research students in the Department, but the majority are given by outside speakers. Additional ad-hoc seminars may also take place.

In statistics, there is an applied statistics seminar where people from outside are invited to present their research work. In addition, staff from statistics report about their research progress in this seminar. The seminar takes place on Wednesdays at 2pm. Research students in statistics are expected to join for this seminar. In addition, there is a seminar for statistics PhD students in which students present their progress regularly.

Meteorology

There are *two* types of *Departmental Seminars* that all research students are expected to attend. Seminars are given by *external* speakers at **midday on Mondays during term times**. Seminars are given by *internal* speakers (including Postgraduate Research students) at 1 pm on Tuesdays during term-time. *Weather and climate discussion* (WCD) runs at midday on Fridays and covers the latest events in the atmosphere and oceans as they unfold. All the *research groups* run their own group meetings (usually weekly) and you are encouraged to attend and participate in any relevant group meetings. Extra seminars with external speakers are often organized by the groups and are advertised by email.

2.7 Demonstrating

Many Postgraduate Research students act as paid demonstrators to support teaching activities within the School. ***You must be eligible to hold paid employment in the UK***. Typically all EU students are eligible, but non-EU students must check the terms of their entry visa to the UK.

Students receive £13.72 (tax free in 2014/15) for demonstrating modules, an amount set by the University. The pay for demonstrating is determined by multiplying this hourly rate by the number of contact hours assigned to the module by the module convener. The rate includes an allowance for the additional time spent in preparation before contact hours. *You can expect to work twice as many hours as the specified contact hours*. Marking opportunities are also available for a limited number of modules. While demonstrating is of course optional and not part of the Postgraduate Research course (although the required training courses are), it does give postgraduate students valuable experience in communication skills.

In *Mathematics and Statistics* demonstrating normally involves supervising undergraduate or MMath lectures or tutorial classes and marking and giving tutorials for groups of about ten second or third year students taking mathematics. Most PhD students will have the opportunity to be involved. For example, demonstrators in mathematics can expect to take at least one group and the usual load is one or two hours marking plus one hour tutorial per week. Interested students should contact the Director of Undergraduate Studies (currently Dr Peter Chamberlain for Mathematics and Statistics modules) at the beginning of the first term. It is mandatory for all mathematics and statistics PhD demonstrators to attend the Teaching Assistant course arranged by the Department which usually lasts for a couple of hours. If you would like to work in the department, please bring your Passport and NI number and see Ruth Harris to set up a zero hours contract

In *Meteorology*, demonstrators are required for tutorial and problem classes and synoptic, instrument, fluids, and computing laboratory classes. A full load of demonstrating is currently around 60 hours per year. Demonstrating can provide a break from research and a useful addition to a C.V. It is **mandatory** for all demonstrators in labs, as well as for postgraduates doing lab/field work as part of their research project, to attend one of the Lab Safety Courses for Postgraduates. We **strongly** recommend that *all new Postgraduate Research students attend one of these courses as failure to do so will significantly limit the demonstrating available to them*. Students will be advised of the dates of these courses. Some non-module demonstrating roles are also on offer such as assistant librarian and coffee monitors. They are currently paid at half the demonstrating rate because they do not require additional preparation time or MSc-level expertise.
Demonstrators will need to have a *zero hours contract*, to arrange for this please contact Zala Hussain (a.hussain@reading.ac.uk). Module conveners determine the duties before the start of the module and job descriptions are available on the demonstrating webpage. You claim for payment after completing your demonstrating role by filling in an electronic claim form. You can obtain the form from Zala Hussain (a.hussain@reading.ac.uk) and you must send it to the module convenor to confirm the hours worked; the module convenor will then send the form to Zala.

2.8 Supervision

You will have at least two supervisors. The general role of the supervisor can be found in the "The Student Journey" at http://www.reading.ac.uk/graduateschool/currentstudents/gs-student-journey.aspx
The success of this relationship can be very important to the success of the degree and is explained in *'The Graduate School Guide to You and Your Supervisor'*
http://www.reading.ac.uk/web/FILES/graduateschool/GSG_YouAndYourSupervisor.pdf. You need to find a working relationship that is mutually acceptable. A regular meeting schedule is a good idea - *weekly* is about right for many students. Your supervisor not only guides your work but also recommends training and travel.

Students who are on a CASE studentship or who have other industrial support will usually meet with their industrial supervisor within the first month. This practice will be continued with regular meetings throughout all three years of research. In addition it is usual that such students will spend periods of time (about a month per year) at the industrial sponsor's establishment. In the spring term of each academic year you will be required to complete an `Evaluation of supervisory arrangements' form. The purpose of this form is to ensure that progress is satisfactory, that you have no concerns regarding supervision or the facilities for research. Students will receive a request to complete this form from the Graduate School. The Dean of the Graduate School will review the forms and take appropriate action.

2.9 Media Engagement Policy

The ability to 'communicate one's science' in a professional, persuasive and articulate manner is more important than ever in today's media-driven world. Media engagement directly includes activities such as giving interviews on local radio, television interviews, and discussing recent weather/climate events with journalists. Related interactions are any in which your opinions are identified by your name and affiliation in an arena typically accessed by the wider public, such as blogs.

The Department of Meteorology has the highest media presence of any department within the University, and recognises that media engagement by PhD students can be a benefit to both the student and the Department/University as well as providing a service to society. However, we also recognise that students often have little or no media experience or training, and as a result can be especially vulnerable to unintended consequences of media interactions. These can lead to an adverse impact on both the reputation of the student and University.

Atmospheric and climate science can attract external interest, for example from the press and wider public. If you are approached directly by the media for comment on your work, your supervisor and the University Press Office should be informed: before commenting, seek advice and, if appropriate, attend media coaching beforehand. (The Department runs a training series associated with media engagement.) However, media engagement is rarely required as part of a PhD, and students are entitled to refuse media requests by the University Press Office, and are strongly advised to do so on subjects where they feel ill-qualified to contribute This is important as the reputations of the University, Department and students and supervisors as individuals can be inextricably linked to such comments.

3. MONITORING AND PROGRESSION

3.1 School Monitoring Procedure

The purpose of monitoring is to ensure that research students are making satisfactory progress and are receiving appropriate levels of supervision and support. Monitoring forms the basis of the decision at the end of the second year that **a student's registration be confirmed as PhD or, exceptionally, changed to MPhil registration**. This decision is passed on to the Faculty for ratification by the Postgraduate Committee. An MPhil thesis should contain results expected of two years research while a PhD thesis must reflect three years of research. Both are expected to make an original contribution to knowledge, but at a higher level in the case of PhD.

The School Director of Postgraduate Research Studies has overall responsibility for the monitoring process. The School monitoring procedure is outlined below. Local practices may differ slightly within these guidelines. All research students should have their progress monitored biannually, in May/June and November/December for full-time students first registering in October and at corresponding intervals in other cases. Intermediate monitoring points may occur routinely or exceptionally on the recommendation of the supervisor(s), monitoring committee or at the instigation of the assessors. In particular, an increase in the rate of monitoring may be appropriate for students whose submission is delayed into their fourth year.

At the end of each academic year the School Postgraduate Office must inform the University Doctoral Research office whether progress is satisfactory for each student. At the end of the second year the list will include a recommendation that a student's registration be confirmed as PhD or transferred to MPhil.

The monitoring of each research student will be carried out by a Monitoring Committee consisting of two assessors, at least one of whom is familiar wherever possible with the relevant research area appointed by the Director of Postgraduate Research, having knowledge of the subject area. For students whose research topic bridges two subject areas there will normally be one assessor from each area; in such cases, the relevant Directors of Postgraduate Research Studies will collaborate in the appointment of the Monitoring Committee. Students will be informed of the composition of their Monitoring Committee and the timing of the first monitoring meeting shortly after registering for their PhD. The School Postgraduate Office will be informed of the composition of each Monitoring Committee.

Supervisors may attend the monitoring meetings at the invitation of the assessors; however, **they must not be present for the whole of each monitoring meeting.** In particular, they **must not be present when the monitoring report is completed** and when the **decision is made by the assessors to recommend that a student's registration is confirmed as PhD or transferred to MPhil**. **The student must be given the opportunity to comment on supervisory arrangements in the absence of his or her supervisors.**

It **is the student's responsibility to arrange a suitable meeting time and room with all the committee members**. Schedule an hour for the meeting, plus 30 minutes for the monitoring committee members afterwards. It can be difficult to arrange a date with the whole committee so please start arranging several months in advance. A week before each monitoring meeting you must circulate a report to all committee members. This is **usually 6 sides of A4 in length in Meteorology and in Mathematics and Statistics, follow the guidelines for the year that you are in**. This report describes your work to date (updating on the previous report and so concentrating on the previous six months), its relation to research in the literature, the aims of your PhD project and plans for its continuation. You should end with a section on the development of your generic and subject specific research skills (often entitled 'transferable skills') and your references list. It is useful to focus on both immediate aims following the meeting (the next few months) and your ambitions for the longer term. In the second year you should start to include thesis plans (i.e. descriptions of chapters and their contents); many students also include time plans. During your *first year* many of the ideas or directions may have been prompted by your supervisor, but as you progress the committee will be looking for evidence that you are developing independence in your research. Don't be afraid to put forward speculative ideas for the longer term – the meeting is a chance to get some feedback from the committee members on which directions might yield more fruitful results. Ask second or third year students to see their previous reports to gauge report content and format. You must also include an updated training record as an appendix to each report – this list will grow with time through the PhD. RRDP courses attended can be downloaded from RISIS and you should supplement this list with any other courses attended. The list should simply include the date, title and type of course (e.g. RRDP ECWMF etc.). The list will enable the committee to spot training gaps and should also be useful to you both during your PhD and for your CV writing. This training record is designed to supplement but not replace the 'transferable skills' section of the monitoring committee report since this section will also include other items such as conference attendance, self-study etc.

Your supervisor must also circulate a report before the meeting, about half a page in length, giving an assessment of your progress and capabilities – you will receive this report.

At the start of each meeting, the Chair will remind everyone of the purpose of monitoring and run through your previous committee report and actions arising from it. Then you can expect to be asked to lead the committee through your report responding to questions along the way. Also don't be taken aback if you are asked more general questions about your discipline. You can regard a PhD as an apprenticeship in research in which you develop a wider view of your subject and its relation to others to the level where you can identify unexplored territory and formulate research questions with proposed routes to attempt to answer them. The goal is not just a narrow focus on answering specific research questions. Questioning in the viva examination and monitoring committees will reflect this. The monitoring meeting is very much two-way: use it to get advice from the committee members or to raise any issues with supervision and the research environment of the School or University.

At the end of each monitoring meeting, a Report of the student's progress will be summarised on a *proforma*. The student will sign the Report and deliver copies of it to all committee members and the **original to the School Postgraduate Office**, together with copies of the student and supervisor reports. The Report will state the timing of the next monitoring meeting.

Where progress is unsatisfactory, copies of the monitoring Report and the reports provided by the student and supervisor(s) will also be given to the relevant Director of Postgraduate Research Studies, who is responsible for taking any action that may be required. Issues that cannot be resolved at local level will be brought to the attention of the School Director of Postgraduate Research.

3.2 Local variations/interpretations:

Meteorology

The first meeting (typically December) report for students registered in October that have taken three courses in the autumn term may be shorter than 6 pages. Students exempted from MSc module assessment will be asked to produce a more detailed report of up to 10 sides for the 2nd committee meeting at the end of their first year. It is expected that the time afforded by not attending modules will have been used to advance your PhD research significantly and the extra space can be used to expand on your research and proposal for its continuation. Note that this the report is not a dissertation and a complete literature review and description of your research is not achievable in the space. Its purpose is to demonstrate that you have the capability to complete your PhD, that you are developing research ideas and ways of following them through in a scientifically rigorous manner.

Progress to second year registration will be subject to satisfactory assessment in the M.Sc. modules (if you took any) and on the basis of your research progress, as evidenced from your monitoring report and your supervisor's report. You must also have attended sufficient generic training courses (see Training section).

A monitoring committee meeting should also take place during the min/max registration year. Students should produce a general timetable of various stages they need to reach before their submission date and it would also be an opportunity to discuss any problems that may have arisen. This could be discussed either via a meeting or via email/skype, but a report form or a statement from the Chair needs to be produced and kept on file to confirm that progress is on target and submission can be achieved.

Mathematics and Statistics

Annual guidelines for writing your Monitoring report are available from the Postgraduate Administrator or the Departmental Director for Postgraduate Studies. Please make sure you that you check for an updated version before your monitoring.

3.3 Timeouts e.g. Suspension, Maternity leave etc.

Illness, financial problems or planned periods of study distinct from your normal research work may necessitate that you arrange for a formal suspension of your studentship. This means that, for a specific period, payment of your studentship is suspended, and restarted when you begin work again such that you still get at least full 3 years of financial support (possibly longer if research council funded). The end date of your registration is changed by the same period. This must be agreed formally with the University (through submission of a form signed by the School Director for Postgraduate Research Studies to the Doctoral Research Office) and funding body and a supporting case made with, for example, Doctor's certificates. Short periods of illness of less than, say, one month do not necessitate such action but cumulative periods may do so. If you are advised not to work for a substantial period (several weeks or more) please obtain a Doctor's certificate to this effect and file a copy of this with the relevant Departmental Postgraduate Office. For some funding arrangements, e.g. the NERC Doctoral Training Program, this evidence may be used to extend your funding. Keeping your supervisor informed as soon as you become ill is important.

Your studentship will also be suspended to cover parental leave. Arrangements for support vary with funding body. For example, studentships funded by UK Research Councils provide 26 weeks maternity leave at the full stipend plus the option of 26 weeks additional leave without pay. The end date of registration will be extended accordingly. The University matches these terms in the studentships that it funds directly.

3.4 Troubleshooting

Pursuing a PhD is rarely a smooth passage to your goal, though we try hard to make it so. Sustained self-motivation when, as it sometimes may seem, nobody understands or cares what you are doing can be a struggle. Your Supervisor, Monitoring Committee and Departmental and School Directors of Postgraduate Research Studies are all there to help so make use of them. If your relationship with your Supervisor breaks down, let the Departmental or School Director of Postgraduate Research Studies know as soon as possible so that solutions can be sought. It is possible to have a new Supervisor assigned in exceptional circumstances. Similarly if you have any serious grievance about the way you are being treated in your department approach either of them and, if necessary, your complaint will be referred to the School or Faculty level (see The University Code of Practice on Research Students, section 9).

3.5 Thesis submission

At the end of each year the department reports on your progress to the University and some details to the relevant research council or funding body. This should give you a good means of judging how close you are to completion. Most full-time PhD studentships are for three or three and a half years and you should focus your energies on submitting your thesis in this time. Often students run over into a fourth year. Please see the section on extensions. **There is a hard deadline at the end of 4 years for full time students**. Your registration as a full-time PhD student expires at the end of 4 years. Furthermore, the research councils impose penalties restricting further studentships for departments whose students take over four years for submission of their thesis.

It is possible to register as a part-time PhD student with a minimum registration of 4 years and maximum of 6 years (see Code of Practice on Research Students).

As stated in the University's leaflet *Guidelines for the Submission of Theses for Higher Degrees*, you must send a letter giving notice of intention to submit the thesis to the Registrar, via the exams office. This must be done several months prior to that in which you intend to submit and the deadlines for giving notice can be found in the University `Rules for the Submission of Theses for Higher Degrees' at http://www.reading.ac.uk/graduateschool/currentstudents/gs-policies-and-procedures.aspx. The thesis should be submitted soft-bound and in triplicate. The *oral examination* (also known as the *viva*) will ordinarily take place within 3 months after submission. The *viva* is a closed meeting lasting several hours involving 3 people: yourself, the internal examiner (from the University of Reading) and external examiner. Occasionally an Independent Chairperson is also present. The outcome of the *viva* may involve corrections that have to be made to the thesis before it is acceptable. Once the examiners have agreed that the corrected thesis is acceptable, it is submitted finally hard-bound. Making and approving the corrections and binding the thesis can take some time and you should not book flights home, etc., before these matters are completed.

For doctoral researchers at the University of Reading, it is now a requirement that an ***electronic copy of your thesis*** be deposited via an approved, secure method. You will be required to deposit an electronic copy of the final version of your thesis into the University's digital Institutional Repository, **CentAUR**. A compulsory training module ('Creating your electronic thesis') has been developed to help guide you through the process of electronic deposition of theses. For more details on electronic deposition and the training course see
http://www.reading.ac.uk/graduateschool/currentstudents/gs-etheseshome.aspx

Mathematics and Statistics

The financial obligation of students submitting their thesis is as follows: *Students are responsible for the cost of preparing and binding their own thesis. This includes copies produced by laser printer and subsequent photocopies.*

Meteorology

It is generally recognized that the *costs incurred in the production of the thesis must be covered by the student*; those involving the research will be met by the Department. In particular, the costs of binding the thesis must be met by the student. The School Postgraduate Office can give advice on binding.

3.6 Extensions beyond three years

NERC offers funding for PhD students to an average of 3.5 years. The extra funding covers maintenance and tuition fees. Students are ***not automatically entitled*** to the extra funding beyond 3 years associated with their studentship and this can't be awarded if the end date of your grant has been reached. To allocate the extra funding Departments apply the following procedure:

In the May/June monitoring committee meeting, monitoring committees are asked to consider whether a 3rd year NERC funded student should be considered for the extra funding. The Chair of the committee should submit a form (web page http://www.met.reading.ac.uk/intranet/staff/student_admin.html) which indicates the case for additional funds based on the criteria set out by NERC and the number of months requested (etc.) to the Director of Postgraduate Research Studies. The Postgraduate Research Board of Studies will consider the cases and make the final decision. The criteria for deciding whether extension funding should be given are attached to every monitoring committee form.

Note that the extension funding can cover time to write journal papers from the PhD research after the thesis has been submitted. Students should discuss with their supervisors early in their third year whether they will try to write papers before submitting their thesis or afterwards. There are pros and cons either way and the best route depends on the nature of the research being conducted (e.g., whether some parts are self-contained and lend themselves to a stand-alone paper). When making this decision it is important to remember that the **thesis must be submitted within 4 years** for fulltime students so you must make a detailed and realistic time plan to be discussed by the monitoring committee.

Any student who has not submitted by the end of their funding (either after 3 years or following the end of their NERC funding extension) must pay a **Continuation Fee** to the University (of about £400-500 for the full year). If you submit before the end of second term of your 4th year, a *partial refund* of this fee can be obtained following submission.

Meteorology

Note that the Meteorology Department may be able to provide some extension funding for non-NERC students funded directly by the Department or University. You should discuss the grounds for extension in the last monitoring committee meeting of your third year and the Chair should write a case for extension (a form can be found on webpage http://www.met.reading.ac.uk/intranet/staff/student_admin.html) to the Director of Postgraduate Research Studies, referring to the same criteria as for NERC. The request will be considered by the Postgraduate Research Board of Studies and an extension granted if the criteria are met and Department funds are available.

The Department also offers 'paper writing' and 'teaching placement' schemes for other non-NERC funded students which can be used following submission. The procedure for application for these extension funds is via a request from the Chair of the monitoring committee, just as for NERC extensions. The monthly payments will match those paid under the NERC extension funding scheme.

4. ANNEX: Use and publication of Information provided to RCUK on RCUK funded Studentships

Briefing for Students, Supervisors and Project Partners

1. PhD project information displayed on the Gateway to Research?

The Gateway to Research (GtR) is a web-based portal http://gtr.rcuk.ac.uk/ where information about funded research is published. The aim is to assist businesses and other interested parties to identify potential partners in research organisations to develop and commercialise knowledge, and thereby increase the impact of publicly funded research. It provides better access for the research community, business and the public to information on research funded by the seven Research Councils and the Innovate UK.

The PhD project information which the Research Councils will publish on the GtR website is given below. Note that the project summary (abstract) is a key piece of content for display in GtR and it must be suitable for publication and not contain sensitive or confidential information.

Item of data	Notes
Student Name	for students starting from 2015 onwards
Training Grants	The grant(s) from which the student is funded. A student may be funded by more than one grant. These are already published on GtR.
Organisation	The organisation that holds the training grant.
Project Title	This should be as informative as possible, even if final title not yet confirmed
Summary	Sensitive or confidential information should NOT be included in this summary
Supervisor	The academic supervisor(s)
Organisation	This will be the Organisation where the student is registered.
Department	The Department of the Organisation at which the student is registered.
Project Partner Organisation	This will be displayed to highlight collaborative working
Registration Date	The date on which the student started their studies
Expected Submission Date	The date by which the thesis is due to be submitted.

2. Other use of information provided to RCUK

Use of submitted data may include:

- Registration and processing of proposals;
- Operation of grants processing and management information systems;
- Preparation of material for use by reviewers and peer review panels;
- Administration, investigation and review of grant proposals;
- Sharing proposal information on a strictly confidential basis with other funding organisations
- To seek contributions to the funding of proposals
- Statistical analysis in relation to the evaluation of postgraduate training trends
- Policy and strategy studies.
- Meeting the Research Councils' obligations for public accountability and the dissemination of information.
- Making it available on the Research Council's web site and other publicly available databases, and in reports, documents and mailing lists.

The following information about training grants and funded students will routinely be made publicly available:

- Student name (for students starting from 2015 onwards)
- Aggregated information regarding student numbers, stipend levels, qualifications, age at start, migration levels (from first degree university to another) etc.
- Name(s) of project partner organisations and supervisors
- Project titles and topics
- Project summaries
- Numbers of students in particular regions, universities or departments in context of the Training Grant funding announced.
- Registration and expected submission dates and rates

Information may be retained, after completion of the Masters or PhD, for policy studies involving analyses of trends in postgraduate training and reporting on these to government bodies such as DBIS. Students should always have been informed that the university is releasing personal details to AHRC, BBSRC, EPSRC, ESRC, MRC, NERC or STFC for the above purposes.

3. Je-S Student Details Functionality

The Research Organisation provides standard information on the details of students and the student research projects funded by the Research Councils' through the web-based data collection functionality Je-S Student Details which Research Organisations use to return details of the students and student research projects funded from the Training Grant.

The Information that is required is available in the Je-S system help text: Go to the following web address, click Show, select Studentship Details and then select Data Protection
https://je-s.rcuk.ac.uk/handbook/pages/StudentResearcherDetails/StudentResearcherDetails.htm